

Single-Peak Crow–Kimura Dynamics: Complete Finite-Genome Spectrum and Exact Projective Gap

Route-A source-local certificate HCS-C336

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Abstract

For the continuous-time mutation–selection equation on binary strings of length L , with symmetric point mutation and one master-sequence fitness spike, we identify the nonlinear simplex flow exactly with a projectivized positive linear semigroup. Walsh decomposition shows that each Hamming eigenvalue retains an explicit codimension-one multiplicity, while all remaining eigenvalues are the roots of one degree- $(L + 1)$ secular polynomial. The secular roots are simple and strictly interlace the mutation poles. The two largest roots therefore give the exact projective spectral gap, with a sharp generic convergence law. Zero selection, zero mutation and the one-locus model are closed separately; finite-length analyticity is not promoted to an infinite-genome error threshold. Exact computation is used only as an implementation receipt.

1 Frozen mutation–selection owner

Let $\mathcal{H}_L = \mathbb{R}^{\{0,1\}^L}$, let F_i flip bit i , and let e_0 denote the all-zero genotype. For $L \geq 1$ and $U, s > 0$ define

$$M_L = \frac{U}{L} \sum_{i=1}^L (F_i - I), \quad A_L = M_L + s e_0 e_0^\top. \quad (1)$$

Both matrices are real symmetric and $1^\top M_L = 0$. On the probability simplex the Crow–Kimura equation in this normalization is

$$\dot{p} = A_L p - (1^\top A_L p) p = A_L p - s p_0 p. \quad (2)$$

Theorem 1 (Exact projectivization and global limit). *For every probability vector p^{in} ,*

$$p(t) = \frac{e^{tA_L} p^{\text{in}}}{1^\top e^{tA_L} p^{\text{in}}} \quad (3)$$

is the unique solution of (2). The denominator is positive, the simplex is preserved, and $p(t)$ converges to the uniquely normalized positive Perron eigenvector of A_L .

Proof. Write $q(t) = e^{tA_L} p^{\text{in}}$ and $Z(t) = 1^\top q(t)$. Positivity of the irreducible hypercube semigroup gives $q(t) > 0$ for $t > 0$, hence $Z(t) > 0$. Moreover $Z' = 1^\top A_L q = s q_0$. Differentiating q/Z gives exactly (2). The same calculation shows that its entries sum to one. Since A_L is irreducible Metzler and symmetric, its largest eigenvalue is simple and has a strictly positive eigenvector. The Perron coefficient of every nonzero nonnegative initial vector is positive; dividing the spectral expansion by $Z(t)$ proves the limit. $\square$

The phrase *projective semigroup owner* records the theorem content of the original manuscript round.

2 Complete finite-genome spectrum

For $S \subseteq \{1, \dots, L\}$ let

$$\phi_S(x) = 2^{-L/2} (-1)^{\sum_{i \in S} x_i}.$$

These Walsh characters form an orthonormal basis and

$$M_L \phi_S = d_{|S|} \phi_S, \quad d_k = -\frac{2Uk}{L}, \quad \langle e_0, \phi_S \rangle = 2^{-L/2}. \quad (4)$$

Put $m_k = L!/[k!(L-k)!]$, $w_k = m_k/2^L$.

Theorem 2 (Full multiplicity and secular factorization). *The complete characteristic polynomial of A_L is*

$$\det(\lambda I - A_L) = q_L(\lambda) \prod_{k=0}^L (\lambda - d_k)^{m_k - 1}, \quad (5)$$

$$q_L(\lambda) = \prod_{k=0}^L (\lambda - d_k) - s \sum_{k=0}^L w_k \prod_{j \neq k} (\lambda - d_j). \quad (6)$$

Thus d_k is retained with multiplicity $m_k - 1$ and the remaining $L + 1$ eigenvalues are precisely the roots, away from the poles, of

$$1 = s \sum_{k=0}^L \frac{w_k}{\lambda - d_k}. \quad (7)$$

The multiplicities in (5) sum to 2^L .

Proof. Let $E_k = \text{span}\{\phi_S : |S| = k\}$. The subspace of E_k whose Walsh coefficients sum to zero is orthogonal to e_0 and has dimension $m_k - 1$; A_L acts there by d_k . Its orthogonal complement over all k has basis

$$u_k = m_k^{-1/2} \sum_{|S|=k} \phi_S, \quad \langle e_0, u_k \rangle = \sqrt{w_k}.$$

The restriction of A_L is $D + s v v^\top$, where $D = \text{diag}(d_0, \dots, d_L)$ and $v_k = \sqrt{w_k}$. The matrix determinant lemma gives (6); adjoining the retained orthogonal spaces gives (5). $\square$

3 Strict interlacing and projective rate

Order the mutation poles as $d_L < \dots < d_1 < d_0 = 0$ and set $R(\lambda) = s \sum_k w_k / (\lambda - d_k)$. On every pole-free interval,

$$R'(\lambda) = -s \sum_{k=0}^L \frac{w_k}{(\lambda - d_k)^2} < 0. \quad (8)$$

At the left and right ends of each internal gap, R has limits $+\infty$ and $-\infty$, respectively. Above d_0 , it decreases from $+\infty$ to 0; below d_L it is negative.

Theorem 3 (Interlacing and exact projective gap). *The $L + 1$ secular roots $\rho_0 > \rho_1 > \dots > \rho_L$ are simple and obey*

$$\rho_0 > d_0 = 0 > \rho_1 > d_1 > \rho_2 > \dots > d_{L-1} > \rho_L > d_L. \quad (9)$$

In particular, the two largest eigenvalues of the full 2^L -dimensional matrix are ρ_0 and ρ_1 . With

$$\Delta_L(U, s) = \rho_0 - \rho_1, \quad (10)$$

every simplex orbit satisfies $p(t) - p_\infty = O(e^{-\Delta_L t})$. This exponent is attained whenever the coefficient of the ρ_1 eigenvector in the projective expansion is nonzero.

Proof. Equation (8) and the one-sided pole limits give exactly one root in every internal gap, one above d_0 , and none below d_L . The retained eigenvalue nearest the top is $d_1 < \rho_1$, so ρ_1 is the true second eigenvalue. Divide the real-symmetric spectral expansion in Theorem 1 by its Perron term. All remaining exponents are at most $\rho_1 - \rho_0$; a nonzero ρ_1 coefficient gives the stated leading term. $\square$

The phrase *strict secular interlacing* records the substantive Round-1 addition.

4 Reducible faces and evidence boundary

If $s = 0$, selection disappears and d_k has its full multiplicity m_k ; the mutation semigroup converges to the uniform law. If $U = 0$, writing $a(t) = p_0(t)$ gives $\dot{a} = sa(1 - a)$. Initial data with $a(0) > 0$ converge to e_0 , whereas the whole face $a(0) = 0$ is stationary. Thus no irreducible Perron conclusion is smuggled onto that face. For $L = 1$, no d_k is retained and direct diagonalization gives

$$\lambda_\pm = \frac{s - 2U \pm \sqrt{s^2 + 4U^2}}{2}. \quad (11)$$

The exact evidence enumerates rational parameter fixtures, reconstructs the factorization independently, directly compares small hypercube matrices, counts secular roots symbolically, checks Walsh eigenvectors and quotient derivatives, and attacks signs, weights, multiplicities, scopes and parser conventions. These finite checks are regression receipts, not an extrapolation proof. At every fixed finite L the strictly interlacing roots vary smoothly in the positive chamber; this paper makes no singular infinite-genome error- threshold claim.

5 Ownership and Route-A boundary

Crow–Kimura mutation–selection theory and the permutation-invariant linear reduction are historical source owners. Nearby workspace packages treat the mutation-only Ehrenfest cube, Moran fixation, Wright–Fisher diffusion and network SIS; none contains the single-peak rank-one full spectrum proved here. The phrase *finite-length boundary and route firewall* records the substantive Round-2 addition.

The Hamming cube supplies neither rational-prime local data nor a logarithmic prime clock. This positive normalized flow has no isolated arithmetic primitive ledger, and (5) is a source characteristic polynomial, not a target Euler factor or divisor. Real symmetry gives only a formal source-operator hint. Under scope NO_BAD_EULER_OR_ROOT_NUMBER, the tuple is

(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT).

The verdict is ROUTE_A_REJECTED; Route B is locked. No target arithmetic datum, Euler factor, root number, automorphy, target functional equation or zero match, or Hilbert–Polya operator is claimed.